

1. Administrative & Study Identification

Full Study Title: An Open-label, Multi-center, Phase IV, Rollover Study for Patients With Sickle Cell Disease Who Have Completed a Prior Novartis-Sponsored Crizanlizumab Study **Short Title/Acronym:** Crizanlizumab Rollover Study for SCD **Protocol Number:** CSEG101A2401B **NCT Number:** NCT04657822 **Posting ID:** 257497740142056145 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Crizanlizumab (Trade Name: Adakveo; Concentrate for solution for infusion for Intravenous use) **Phase of Development:** Phase IV **Trial Status:** Recruiting **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To provide continued access to crizanlizumab and assess long-term safety and tolerability in patients with Sickle Cell Disease (SCD) who have completed a prior Novartis-sponsored crizanlizumab study. **Secondary Objectives:** To evaluate long-term efficacy (including annualized rates of vaso-occlusive crises [VOCs]), persistence of P-selectin inhibition, and overall safety of continued crizanlizumab administration.

2.2 Background & Rationale This multi-center, multi-national rollover study is designed to allow continued access to crizanlizumab for patients with sickle cell disease (SCD) who are currently on crizanlizumab treatment in a Novartis-sponsored "parent" study and are benefiting from the treatment, as judged by the investigator. The study ensures that patients do not experience a gap in therapy before the medication becomes commercially available or accessible via other local healthcare mechanisms.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A **Trial Status:** Recruiting

3.2 Planned Enrollment & Duration Target \$N\$: 130 participants **Number of Sites:** Multi-center, multi-national (including sites in France, Germany, Italy, Spain, Turkey, Lebanon, and Oman) **Estimated Study Start:** 10-Jun-2021 **Estimated Primary Completion:** 10-Jun-2031

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Written informed consent/assent, according to local guidelines, signed by the adult patients or by parents/legal guardians for patients under 18 years. 2. SCD patient currently enrolled in a Novartis-sponsored study receiving crizanlizumab, has fulfilled all requirements in the parent study, and is currently benefiting from the treatment as determined by the investigator. 3. Patient has demonstrated compliance to the planned visit schedule in the parent study and has shown willingness and ability to comply with future visit schedules.

4.2 Exclusion Criteria 1. Patient had permanently discontinued from crizanlizumab study treatment in the parent study before the parent study completion. 2. Ongoing/unresolved treatment-related Grade 3 or higher adverse events (AEs), and/or any ongoing AE requiring dose interruption. 3. Concurrent participation in any other investigational clinical trial other than the parent study. 4. Pregnant or nursing women, and women of childbearing potential unwilling to use highly effective contraceptives. 5. SCD patients who do not meet parent study protocol criteria to continue with crizanlizumab.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Crizanlizumab will be administered at the same dose and schedule as in the parent study (typically 5.0 mg/kg or 7.5 mg/kg administered every 4 weeks). **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** Long-term continued access (up to 10 years, or until commercial availability/reimbursement in the respective indication).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies for SCD as continued from the parent protocol. **Prohibited:** Concurrent participation in any other investigational clinical trials.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	N/A No screening period; direct transfer from parent study
Baseline	Day 0	Rollover consent, Baseline Safety Labs, Resumption of IV Infusion	
Interim	Every 4 Weeks (28 days)	Ongoing Crizanlizumab IV Infusion, AE Monitoring, Efficacy Assessment	End of Study / Safety Follow-up
+105 Days post-last dose	Final Safety Evaluation (Not applicable if transitioning directly to commercial drug)		

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Not Applicable as this protocol is to provide an option for continued access to crizanlizumab for patients with Sickle Cell Disease who have completed a prior Novartis-sponsored Crizanlizumab study: Protocol to provide an option for continued access to crizanlizumab for patients with Sickle Cell Disease who have completed a prior Novartis-sponsored Crizanlizumab study, benefited from the treatment and do not have access to reimbursed, commercially available crizanlizumab.

Measurement Method: Clinical assessment, laboratory monitoring, and standard pharmacovigilance reporting.

7.2 Secondary & Exploratory Endpoints 1. Number of participants with treatment emergent adverse events.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at every 28-day infusion visit. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor and relevant regulatory authorities. **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by Novartis Pharmaceuticals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive at least one dose of crizanlizumab in the rollover study). **Sample Size Justification:** N=130; Sample size is driven by the number of eligible patients rolling over from the parent crizanlizumab trials. **Handling of Missing Data:** Standard multiple imputation/LOCF methodologies consistent with the parent trial's SAP for long-term tracking.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring to ensure rollover continuity and patient safety. **Audit Trail:** Continuous eCRF tracking linked chronologically to the parent study databases.

11. Study Identifiers & Governance

Primary ID: CSEG101A2401B **Registry Number:** NCT04657822 **Posting ID:** 257497740142056145 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Rollover Study for Patients With

Sickle Cell Disease Who Have Completed a Prior Novartis-Sponsored Crizanlizumab Study **Official Regulatory Title:** An Open-label, Multi-center, Phase IV, Rollover Study for Patients With Sickle Cell Disease Who Have Completed a Prior Novartis-Sponsored Crizanlizumab Study

12. Clinical Framework (Sickle Cell Disease Specific)

Primary Condition: Sickle Cell Disease (SCD) / Anemia, Sickle Cell **Diagnosis Threshold:** Confirmed SCD diagnosis and completion of parent crizanlizumab clinical trial **Medical Methodology:** P-selectin inhibition (monoclonal antibody therapy) **Optimization Target:** Prevention and reduction of Vaso-Occlusive Crises (VOCs)

13. Study Procedures & Methodology

Management Pathway: Long-term continued access and observational tracking **Education Protocol (HCP):** Monitoring guidelines for infusion-related reactions **Education Protocol (Patient):** VOC symptom tracking and adherence to the 28-day infusion schedule **Quality Audit Mechanism:** Tracking patient compliance mapped directly to the parent study visit schedule

14. Observation & Follow-Up Schedule

Total Duration: Up to 10 years **Onsite Visit Cadence:** Every 4 weeks (28 days) for IV administration **Remote Monitoring Frequency:** As necessary between infusion windows **Checklist Review Interval:** Safety Follow-Up specifically occurring 105 days after the last dose

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator			[Insert Name]			_____			[DD-MMM-YYYY]							
	Clinical Operations Lead			[Insert Name]			_____			[DD-MMM-YYYY]							
	Lead Statistician			[Insert Name]			_____			[DD-MMM-YYYY]							